



VCU College of Engineering

MULT-26-608

Computer Vision-Based Settling Rate

Analysis for Water Treatment

Project Proposal

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Executive Summary

This project, sponsored by ChemTreat, focuses on developing a precision camera apparatus designed to enable accurate and reliable settling analysis of coagulants. The primary goal is to measure and compare settling times across different coagulants, identifying the most efficient option for any given raw water source. Automating this process addresses longstanding limitations of manual jar testing by improving objectivity, consistency, and efficiency, while also facilitating data driven decision making in water treatment applications. The apparatus is designed to capture high quality video of particle settling while maintaining reproducibility and portability. To date, the majority of the system has been assembled with the exception of the camera arm, which will be integrated in the near-future. In the interim, a precision adjuster and tripod will be used to maintain alignment while data collection proceeds. The computer model is currently producing preliminary outputs which will be refined through further training and accuracy evaluation. The design will emphasize portability, quick setup, and compatibility with data integration in Excel. The key deliverables include the imaging system, custom software, and a labeled dataset in Excel, along with standard operating procedures and a verification report documenting the system's accuracy and reliability. Success will be measured by improved consistency compared to manual jar testing and the system's ability to produce clear, repeatable results. This project aims to bring automation and objectivity to a process that has remained manual for decades, ultimately improving efficiency and usability.

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Section A. Problem Statement

Most industries rely on water treatment in some shape or form, as many processes depend on a consistent supply of high-quality water for cooling, cleaning, chemical reactions, and product formulation. Proper treatment helps prevent equipment damage from scaling, corrosion, and fouling, overall reducing downtime and maintenance costs. It also ensures compliance with environmental regulations, optimized chemical and energy usage, and supports overall operational efficiency^[1].

ChemTreat is a leader in the water treatment industry, designing industrial water treatment solutions for a variety of customers in industries such as healthcare, chemical processing, and food and beverage^[3]. These customers rely on water treatment solutions that enhance system efficiency, reduce operational costs, and ensure compliance with regulatory and quality standards.

The first step of water treatment is the effective removal of suspended solids and other contaminants. To achieve these outcomes, it is crucial to remove contaminants early in the process through effective treatment methods. This is done through the addition of coagulants to bind suspended particles in raw water to form larger clumps, accelerating their settling and removal^[2]. A variety of coagulants are able to accommodate the unique characteristics of different raw water sources and desired treatment outcomes. Because of this, tests must be conducted to identify the best coagulant to suit the unique needs for each water treatment process. This ensures the selected coagulant performs effectively to optimize treatment efficiency and water quality^[4].

Jar testing is used to evaluate the effectiveness of a coagulant in facilitating solid-liquid separation by mimicking a full scale water process in a smaller environment. It does this by varying the type, dosage, and sequence of treatment chemicals across multiple water samples which are then subjected to a treatment process involving coagulation, flocculation, and filtration in order to assess the best coagulant to their system^[4,5]. The current limitation with this technology is that it relies on manual testing procedures and subjective observation, which introduces variability between operators and increases the likelihood of human error. Factors such as individual judgement and procedural inconsistencies can all affect the accuracy and repeatability of results. Additionally, traditional jar testing is typically confined to laboratory settings which limits its responsiveness to changing conditions in the field^[5].

Our project aims to address these limitations by developing a portable jar testing apparatus that uses machine learning and computer vision to objectively measure settling behavior. By replacing subjective visual assessment with quantifiable image-based metrics, the device would provide constant, repeatable data that reduces operator to operator variability. Its portable design would enable testing to be performed in the field, allowing operators to respond more rapidly to fluctuations in water quality and make informed treatment adjustments without waiting for laboratory results. Overall, this system would give ChemTreat faster, objective settling data in the field, enabling them to deliver stronger technical support for their customers.

While ChemTreat is the main client for this project, these solutions will serve a diverse range of stakeholders including industrial water treatment companies, their customers and stakeholders, regulatory agencies, environmental groups, and communities that depend on safe, quality water. By creating a portable method for quantifying solid settling times in water treatment, this technology will support treatment operations and improve adaptability of water systems.

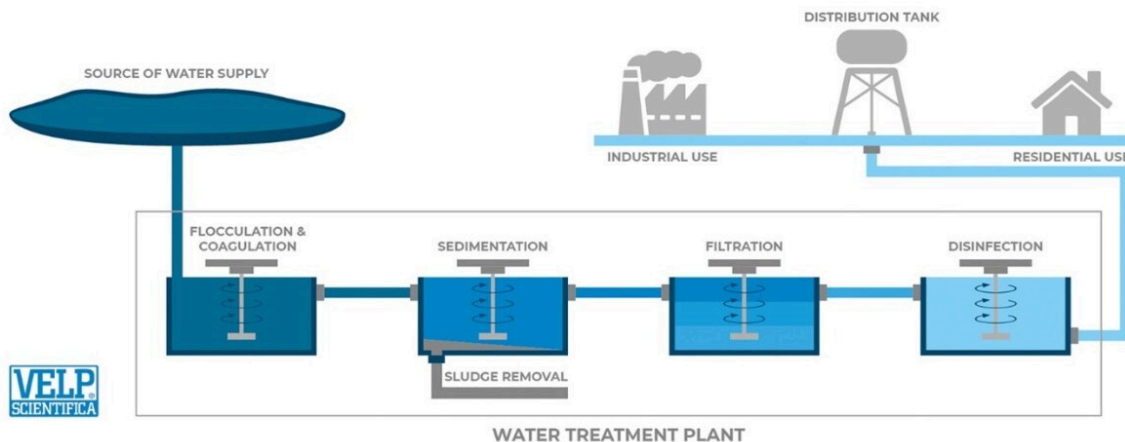


Figure 1. The stages of water treatment from Fisher Scientific Lab Reporter [2].

Previously, another student team designed a prototype around this vision-based learning model. It showed much success in proving concepts, allowing short-term predictions of settling rates. However, there were still some limitations this project seeks to improve upon. For one, the prototype still requires a good amount of setup and does not quite meet the mobility requirements desired by ChemTreat. Secondly, the prototype is only able to predict one water sample at a time, which does not reflect the current industry standard and needs to have comparative testing.

There was a similar study involving machine learning and water treatment conducted in Japan that used convolutional neural networks (CNN) to optimize dosage of polyaluminum chloride, a coagulant. The experiment used a Sony RX1R digital camera to record 30 jar tests for 22 minutes each. Over a million pictures were collected and processed using an NVIDIA Geforce gtx 1080 GPU. While the tests were catered to Japanese water standards, the study shows that machine learning is applicable and beneficial to water treatment^[5].

Other attempts at using computer vision-based learning to predict settling rates have also been used, with similar issues as this project's predecessors. A prototype machine and software, SWAMVA, made lots of headway in using machine learning and vision-based computer technology to accurately and quickly predict solid settling rates, greatly reducing prediction times.^[6] SWARMVA, however, still struggles with mobility, accessibility, and small sample sizing.

Section B. Engineering Design Requirements

B.1 Project Goals

From the client's perspective, the purpose of this project is to improve how settling performance is assessed during jar tests and pilot runs. The client values results that are objective, fast, and easy to compare across chemicals and doses. The goals below are written as outcomes the client needs, not descriptions of the design.

- We want to deliver an objective and repeatable standardized settling measurements to then replace the manual reading of the jar tests.
- We want to make the processes faster and the decision making and the reading of the jar test more consistent.
- We want to be able to provide clear and realtime results that are all met within the safety standards and be within the budget.

B.2 Design Objectives

The overall objective of this product's design is to predict the effectiveness of different treatments for settling suspended solids in water. This design seeks to improve upon the established jar test by creating a setup that is more portable and accessible for researchers who travel to client sites to provide estimations for effective treatments. The system will use machine learning and computer vision tools to build and train a model that can accurately predict settling behavior. Since the settling time of suspended solids typically ranges from 5 to 20 minutes depending on the jar sample, the goal is to predict the final settling time within 3 to 5 minutes, saving up to 15 minutes compared to traditional methods. The design aims to deliver an objective, repeatable, and standardized approach to settling measurements, replacing the subjective visual estimates currently used throughout the industry. This test will allow for more data based decisions, ensuring consistency and comparability across operators, chemicals, doses, and various raw water sources. The system is intended to integrate seamlessly into existing lab workflows while adhering to all safety standards, budget constraints, and project limitations.

B.3 Design Specifications and Constraints

Below is a list of design specifications and constraints that the project will follow:

- Physical constraints
 - 2' x 2' x 2' dimensions
 - Contain more than one jar
 - Symmetrical even amount of jars
 - Consistent or rigid camera rig
 - Constant and bright lighting
- Video recording requirements
 - Video must be recorded with a resolution of at least 1080p
 - Video will be recorded with at least 15 fps
 - Video should be recorded using a smartphone, specifically an iPhone
- Machine learning requirements
 - Device must have at least 8 GB of RAM
 - Device must be at least 256 GB of storage
 - Program must use publicly available machine learning and computer vision libraries
 - Program must be able to analyze between two to four jars side by side
 - The trained model must return a prediction of final settling time within 3-5 minutes of the jar test

B.4 Codes and Standards

Below is the list of specific codes and standards that apply to this project, with a brief note on how each one relates to the design, testing, and operation of the prototype. Standards guide the technical methods and quality expectations. Codes are legal or regulatory requirements that must be met.

1.EPA Method 180.1 – References the method for turbidity used for the ground truth comparison of predicted NTU values in low ranges.

2.ISO 7027-1:2016 – This is the international turbidity standard and the turbometric procedures used to validate low and high turbidity waters.

3.ASTM D2035 –This is the standard practice for coagulation for jar tests and defines the variables dosing steps and the evaluation criteria.

4.Standard Methods 2130 B – This is the lab procedure for turbidity and this is to ensure the collaboration to align with the common lab practices.

5.IEC 61010-1 is the safety requirements for electrical equipment used in the laboratory to measure and control the lights and the camera and finally the rig itself.

Section C. Scope of Work

C.1 Deliverables

The final prototype of this project will have two major components to it: the physical testing rig and the coding and analysis of data collection. For the physical testing rig, it will contain two to three rectangular jars side-by-side so that a video can be collected of all three jars simultaneously. This is to ensure we satisfy the comparative component desired from the project by ChemTreat as it better aligns with timeliness and current jar test designs. These jars should be around 250ml in size to ensure accurate reflection of the sampled water's characteristics and give enough suspended solids presents to be settled. Furthermore, the jars will have an agitator in some form to appropriately mix the water samples collected before settling rates are measured. Furthermore, the physical rig will require a camera, one that should ideally be supplied by the tester in the form of their personal or work smartphone's camera. This meets the mobility and accessibility requirements of our Client, ChemTreat. This implies the need for a built in or supplied camera stabilizer and holder to obtain more accurate data for the treatment analysis. Similarly, lighting will also be an important factor to control for visual-based data collection. Thus, lightstrips should be attached to the physical rig. This to better assess turbidity of the water to then correlate it to the settling rates of the suspended solids in the water sample. The physical rig hopes to best eliminate any disturbances that the video processing might have to work around while maintaining the overall goal of a mobile and accessible prototype.

The data analysis aspect of this project also has to keep the mobility and time frame constraints in mind. The video will be first gathered on a smartphone camera that meets the 1080p and 15 fps requirements. This video will then be sent to a computer more capable of machine learning program execution. This video should then be analyzed and able to return a comparative settling rates analysis within the first five minutes of filming. This will require an exportation of the video's data to a third party program that can then interpret the visual data as it relates the water sample's turbidity, learn expected behaviors, and make settling predictions off that. That returned data should entail a simple and easily digestible comparative analysis of the two to four jars present in the physical rig. That is to say, the coding should return expected settling rate times and effectiveness of that settling for each jar test and compare the jars to one another. This is to quickly gather which coagulant, flocculant or other treatment would be best suited for the sampled water in a timely manner and in a way that most field testers can understand.

Overall the prototype aims to give preliminary settling rate estimates of two to four water samples simultaneously and compare the results. The actual settling rate tests will only need to last some 5 minutes and predict the next 15 to 25 minutes of solid settling behavior in the sample water. After that 5 minute data collection is done, the video used to record should be sent for interpretation and return an easily comprehensible and comparative analysis.

C.2 Milestones

<u>Tasks</u>	<u>Timeframe</u>
Team Contract	8/27/2025 - 9/5/2025
Project Design Proposal	9/5/2025 - 10/10/2025
Apparatus Development	10/7/2025 - Spring Semester
<i>First Prototype *will be completed after setting up the precision adjuster with the tripod</i>	<i>01/16/2026</i>
<i>Final Prototype</i>	<i>Early to Mid February</i>
Model Development	10/31/2025 - Spring Semester
<i>Model Generates Output</i>	<i>12/5/2025</i>
<i>Single Calculation Demo</i>	<i>First meeting back</i>
<i>Model Training</i>	<i>01/12/2025 - March</i>
<i>Analyze Model Accuracy on Settling Curve</i>	<i>01/12/2025 - March</i>
<i>Model Generates Valid Output</i>	<i>End of March/Early April</i>
Data Collection and Training	01/16/2026 - March

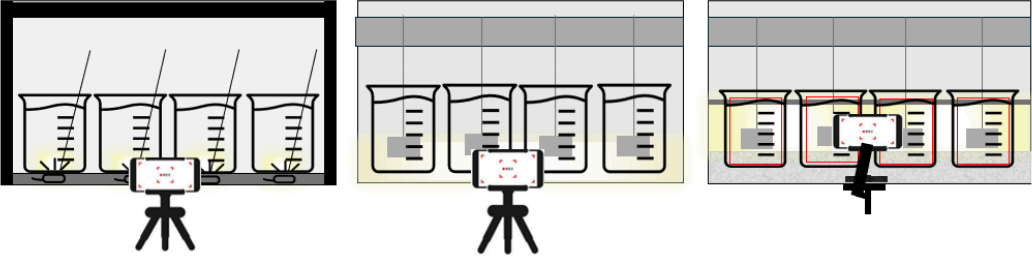
Section D. Concept Generation

The generation of each prototype was rooted in the three major goals of our project as provided by Chemtreat. First being portability, as discussed previously, there is a soft limit to the dimensions of being a 2' by 2' device. Second being the comparative requirement, that is to say expanding the design from one jar to multiple. The third constraint being standardization, or as the physical rig is concerned, consistency.

The first design conceptualized focused on the portability of the design and was an expansion of the previous year's design. It was going to consist of a large photo box and four display lights to go beneath the four jars. A large flaw within this design that made it unviable was the matter of stirring. For a jar test, it is important to have simultaneous and identical stirring to each jar sample to make sure the settling rate times are as a result of the coagulant and flocculants alone. The introduction of the bottom display lights removed the possibility of a magnetic stir bar. Furthermore, a magnetic stir bar runs into its own unique issues. The first being it is an object impeding upon the actual settling of the solids in the jar. Next there was the issue that a stir bar can't reliably mix a larger water sample uniformly, and it would mix differently across each jar. Overall, that stirring was large enough that the idea of creating a machine from the ground up was unideal.

The second iteration of the design focused instead on the comparative design aspect and overall robustness. ChemTreat has an already in use apparatus known as a mini-jar test machine. It meets their portability requirements with the largest dimension being 23". The mini jar test benefits from a stable design that would give consistent results assuming proper experimentation. The stirring mechanism in particular meets the need of being uniform and simultaneous across all four jars. The largest issue with this second design is a similar undiscussed in the first. That issue is the camera holder and positioner. A tripod would be manually set up to put the four jars in frame of the recording. While this would not be an issue during a standard jar test, this project includes the use of machine learning.

Table 1. Decision Matrix.



Design A	Design B	Design C
Pros • Most control over dimensions (portability) • Smaller water samples required	Pros • Most familiar • Cost effective (existing equipment) • Fewest modification and most robust	Pros • Removable modifications • Controls for the most variables • Strongest signal for mean-intensity turbidity
Cons • Least robust • Unfamiliar • Poor mixing (manual)	Cons • Lighting issues • Camera misalignment (poor for model training)	Cons • Least portable • Modified to specific apparatus
Worst	Middle	Best

Section E. Concept Evaluation and Selection

To choose the design that works best for our turbidity system, we used a decision matrix based on the three main options in the figure. Design A, which is the tripod facing the jars, was the most portable but it wasn't very stable and didn't give good consistency. Design B was the setup we were most used to, but it had lighting and alignment problems, so it didn't help with getting clear data. Because of that, we looked at what worked from both and ended up with Design C. It has the most control, the most adjustments, and gives us the cleanest measurements, even though it's not as portable. We can still fix the portability issues as we keep working.

The weights for the criteria came from what the client said mattered most. Accuracy and how well we can track the settling curve were the top priorities, so those got the highest weight. Portability mattered less because it doesn't affect the main goal. Quantification strength came from how stable the camera was and how steady the lighting stayed. Portability was based on how easy it is to move and set up the design. Cost was based on how much new equipment or changes we needed. After scoring everything, Design C came out the highest. It gives the most reliable turbidity readings, the most stable images for the model, and controls the most variables. So Design C is the one we are moving forward with.

Section F. Design Methodology

F.1 Experimental Methods

Evaluating our prototype and making it viable for future data collection involves two major components. That being the viability of the lighting changes and the consistency and robustness of the camera positioner.

The first component, lighting, will be relatively subjective with its acceptance. So long as there are not notable lighting differences between the jars, the lighting will work. The best way to ensure consistent lighting after making the modification from bottom lighting to back lighting is with the use of diffusion fabric. This is to avoid the aggressive lines caused by the strip lighting. With the introduction of the diffusion fabric, there then becomes an issue of folds and shadows. Thus another modification has to be made in the form of a frame to keep the fabric taught and avoid the awkward shadows. After that, the camera footage was sent to the coding team to confirm it would not interfere with data collection. Currently, the lighting tolerance in a subjective measure.

The second process involves the quality of the lighting modification made to ensure the camera is held in a constant enough position to be viable for the autocropping to work consistently and viable for the machine learning program. To accomplish this, numerous trials of the same experiment will be conducted. The goal of these trials is to assess how easy or difficult it is to have a consistent camera angle, and whether or not that affects the data and autocropping. Ideally, it should be reasonable for a human to quickly set up the camera and for the apparatus to do the majority of the aligning by its design. This involves having a fixed height and width position slightly off-center to account for a phone's typical camera position. Overall, there will be some subjectivity to the camera's consistency, but the performance of the code will be used as a more objective frame of reference. Once several trials have been run and confirmed the stability of the camera, more iterations to the coding aspect can be designed.

F.2 Computational Methods (e.g. FEA Modeling, CFD Modeling, etc.)

The computational part is going to be using things like OpenCV to process the video frames and TensorFlow to build the model and develop it so then its going to be able to read the jars. Each frame goes through background subtraction so we can isolate what's happening inside the jar. After that, we look at pixel density and light density across the frame. Pixel density helps us see how the particles shift and settle, and light density helps us check if the lighting stays even. Both of these matter because changes in lighting or pixel distribution can affect how accurate the settling curve is.

Once the frame is cleaned up, we track changes in average pixel intensity over time. This is how we build the settling curve. We assume the camera stays in the same spot and the lighting doesn't move around, because small changes can change the readings a lot. If anything in the physical setup changes during testing, we will update the pipeline so the results stay consistent.

The TensorFlow model will be trained on labeled settling data. It will learn how intensity and pixel density trends relate to turbidity levels. We're not doing advanced simulations like CFD or FEA for this project. The focus is mainly on stable images, consistent lighting, clean frame processing, and pulling out the right trends from the video data.

F.3 Validation Procedure

To validate the design, we are checking two main things. First, the system has to match the design specs we set earlier. Second, it has to actually work the way the client needs it to. For verification, we will compare the output curves to manual readings taken during the jar test. If the shape and timing match well, then the system is working as intended.

We will run the same jar test multiple times with the same conditions to see if the results repeat. If the curves look similar each time, then the system is consistent. We will also vary lighting and camera distance a little to make sure the system doesn't break under normal changes. The goal is to show that our setup can handle routine use without giving bad data.

For validation, we will meet with the client and run the system during an actual jar test they are familiar with. If the settling behavior the system shows matches what they expect from experience, and if the turbidity trend makes sense, then the design is considered valid. This also gives them a chance to tell us if anything is confusing or needs to be changed.

Section G. Results and Design Details

G.1 Experimental Results

Experimental results thus far have been largely exploratory and focused on validating the prototype. Thus far, there have been large successes with the lighting modifications and the small changes necessary to the jars and bottom surface. Moving from the machine's built-in bottom lighting to a modified back lighting proved incredibly beneficial both in adding brightness to the design and video capturing but also avoiding the drastic brightness changes possible as settled solids would block the original light source. The diffusion fabric was confirmed plenty sufficient to soften the bright lights and the lines created by their source.

The rig to position the camera is still undergoing iteration, but significant progress has been made. First, the design requires the camera to be 16" away from the machine to accurately capture all four jars. Secondly, the rig does not need to add any height to the camera once placed level with the bottom of the apparatus. This means so long as there is a rigid structure to hold the camera out attached to the rig, there only needs to include a stable and fixed phone holder to get consistent recording. This should be best achieved using something as simple as wooden plank, though designs have not yet been made with that in mind. On the more experimental side, a couple of preliminary jar tests have been completed, each giving results that will help guide the project later in the confirmation of the machine learning program's effectiveness.

G.2 Modeling Results

For the modeling part, we focused on processing the video frames and checking how well the system could track the settling. After running the OpenCV pipeline, we were able to get clean settling curves by looking at pixel intensity, pixel density, and light density across the frames. The background subtraction worked well once the lighting and the camera stayed consistent. When the camera was stable, the intensity dropped in a smooth curve, which matched what we expected from a real jar test.

We also tested different threshold values to see which ones gave the cleanest signals. Lower thresholds picked up noise, and higher ones cut off important parts of the curve. Once we found the right range, the curves started lining up with what we saw in the physical test. These results showed that the system can pull out the settling behavior without needing any complex simulations.

The TensorFlow model ran on the processed data and was able to read the trends in intensity and pixel density. We didn't push for high-level prediction yet, but the early results showed that the model can follow the settling pattern and respond to changes in turbidity. So the modeling results showed that the design works as long as the camera stays stable and the lighting is controlled.

G.3 Final Design Details/Specifications

The final design is based on the Design C setup. The camera is fixed to the jar test with an adjustable mount so it stays aligned the whole time. The system also has controlled lighting on both sides to keep the brightness even. This helps stop the frame from having shadows or bright spots that mess with the intensity readings.

The setup includes a clear path for the camera, a marked position for each jar, and simple hardware that can be tightened or adjusted without tools. We kept the frame modular so parts can be removed or swapped out during testing. The wiring and power for the lights are kept on one side to avoid glare and reflections.

On the software side, the system runs the same OpenCV pipeline every time. It pulls each frame from the camera, removes the background, checks pixel intensity and density, and builds the settling curve. All of this is saved for the TensorFlow model so it can keep learning from new tests.

Overall, the final design meets the main goals. It gives stable images, clean data for the settling curve, and a setup that stays consistent between runs. It's not the most portable yet, but the performance and accuracy are strong, and that was the main priority.

Section H. Societal Impacts of Design

H.1 Public Health, Safety, and Welfare

The design will help improve water treatment by helping obtain more accurate coagulant dosages. Obtaining more accurate dosages will impact the amount of waste removed from the water treatment process leading to cleaner water.

H.2 Political/Regulatory Impacts

The design will replace manual measurement methods with a more consistent and reliable measurement method. The new design will provide an easier way to compare turbidity between tests.

H.3 Economic Impacts

The design will help reduce material costs and time spent on testing for water treatment. It can reduce the total time spent on jar testing to allow for more time for other tasks. As a result of conducting fewer tests, it reduces the amount of materials spent on testing leading to lower costs. Accurate dosages can affect waste products which is another cost.

H.4 Environmental Impacts

Getting more accurate coagulant dosages will impact sludge production. The type and amount of coagulant used can affect how much sludge is produced so having accurate dosages of coagulant can impact the production of sludge. So getting an accurate coagulant can reduce sludge production.

Section I. Cost Analysis

The min-jar apparatus was provided by our project sponsor, ChemTreat, which significantly reduced initial equipment costs. Because the intended users are expected to have access to an iPhone for data capture and operation, no additional camera costs are required. Under these assumptions, the design modifications implemented for this project remain cost-effective and accessible.

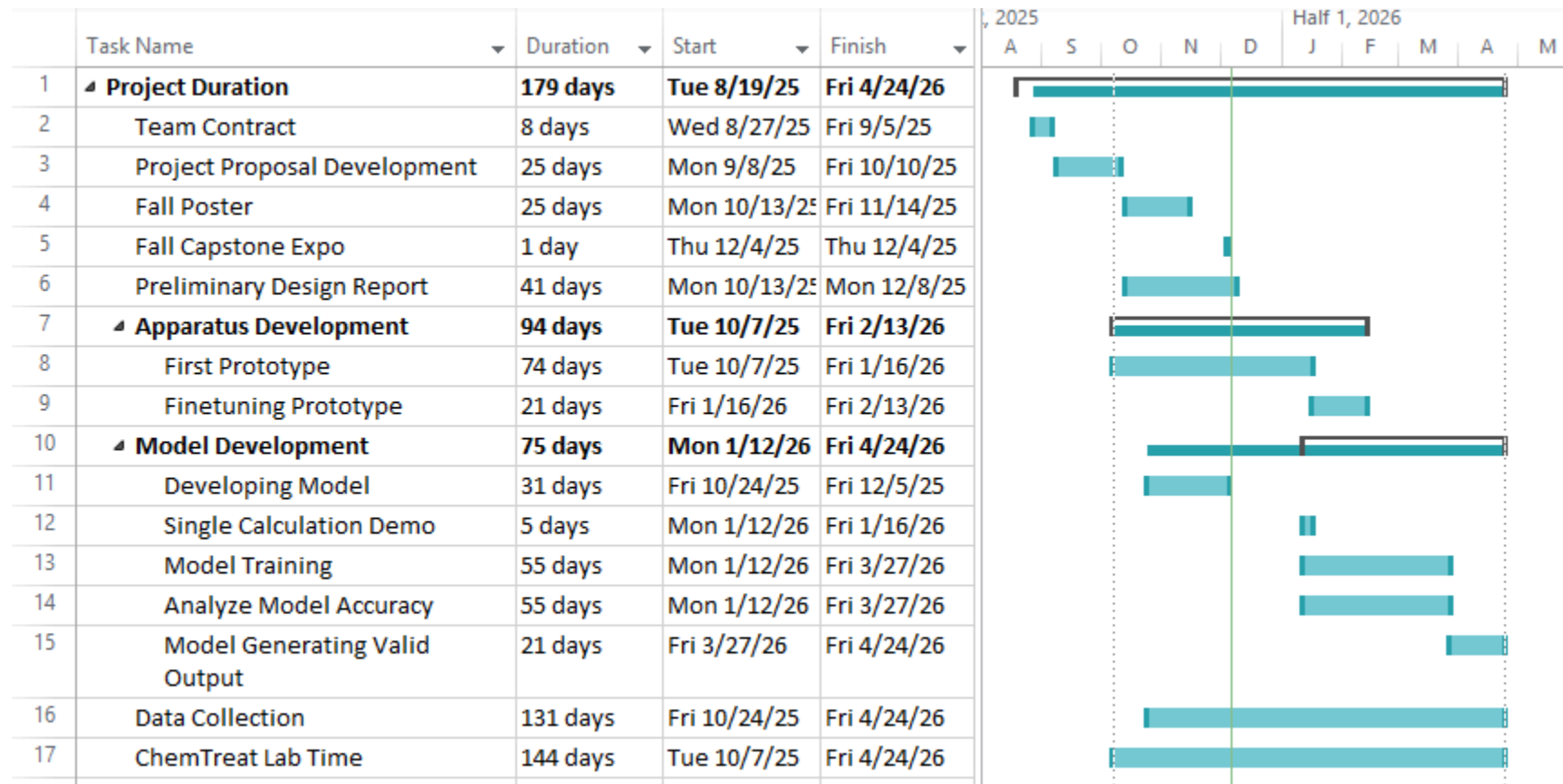
Table 2. Cost Analysis Table.

Equipment		Price
*Mini-Jar Apparatus		\$4800
Lighting Modifications		\$40
Camera Positioning Control		\$30
Jars best for Model		\$25
Surface Modifications		\$25
Camera to computer (data collection)		\$10
*assumed possession of phone camera		*\$700
Total w/o Apparatus	Total w Apparatus	Total w/ Phone
\$130	\$4930	\$5620

Section J. Conclusions and Recommendations

The NovaTech AFM-JM-4 apparatus was modified for this project with a phone stand attached to the apparatus for consistent camera positioning, a raised back stopper to enable backlighting, and a nonslip bottom to improve jar stability. Video is captured at 30 Frames per Second with auto-cropping capabilities, ensuring reliable and precise data collection for dew settling analysis. The majority of the apparatus has been completed, and our next focus will be on integrating and securing the arm. In the meantime, we will be using a tripod, using a string as an optical alignment device to ensure camera position stays secure so we can move forward with our data collection. The computer vision model is operational and currently generating preliminary outputs, but additional experimental videos are required to complete training and assess predictive accuracy. In the future, we will start conducting jar tests to gather data for training and validating the CNN model. We will also refine lighting and calibration curve to ensure differences between jar positions has minimal impact on testing results and tweak the CNN model to obtain more accurate and consistent predictions.

Appendix 1: Project Timeline



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

Team Member Name	Strengths each member bring to the group	Other Info	Contact Info
Aubrey Carey	Lab Research Experience Coding Industry Experience (Process Controls)		careyae2@vcu.edu 703-853-2511
Rohan Sharma	Coding Great Questions! Debugging App Design		sharmar16@vcu.edu 571-831-1762
Virginia Anderson	Communication skills Chemistry Some Water treatment knowledge		andersonvc2@vcu.edu (757)-947-4942
Ronald Menendez	Coding Problem solving skills		menendezrd@vcu.edu (804)-432-6759

Other Stakeholders	Notes	Contact Info
Dr. McGill Dr. Thomson-McInnes	<i>Faculty Advisors</i>	mcgillc2@vcu.edu btmcinnes@vcu.edu
Byamungu Ronald Praise Adebisi	<i>Project Management Advisors</i>	ronaldb@vcu.edu adebisip@vcu.edu
Vlad Djukanovic Will Henderson	<i>Sponsors</i>	vladimird@chemtreat.com williamh@chemtreat.com

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page ([webpage](#) | [PDF](#))

<i>Culture Goals</i>	<i>Actions</i>	<i>Warning Signs</i>
Constructive Feedback	• Set aside time in team meetings to review one another's work and offer suggestions	Editing other's work without consultation
Be proactive	• Setting personal goals of what needs to be done by next group meeting • Doing tasks in parallel and working together to ensure deadlines are met • Not leaving things till the last minute	Missing deadlines Holding up the team Not showing up to meetings
Clear Communication	• Communicate workload via individual and group action items • Organizing files • Ensuring teammates understand the work you're doing	Not responding in text groupchat/discord

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings?
See examples below.

<i>Meeting Participants</i>	<i>Frequency Dates and Times / Locations</i>	<i>Meeting Goals Responsible Party</i>
Students Only	Once a week, on Tuesdays via Online calling and video	Update group on day-to-day challenges and accomplishments Assign or delegate work needed for project to progress Actively work on project
Students + Faculty advisor	Once a week most weeks, on Fridays at 11 am in W101	Update faculty advisor and get answers to our questions Set up/update calendar with hard deadlines
Project Sponsor	TBD Meet in person or by zoom	Update project sponsor and make sure we are on the right track

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, *in addition to* contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my “natural” role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group**, discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for the client/sponsor. This person will schedule meetings, send updates, and ensure deliverables are met.

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
Required: On Edusourced, under the Team tab, make sure that this student is assigned the Project Manager role. This is required so that Capstone program staff can easily identify a single contact person, especially for items like Purchasing and Receiving project supplies.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test

protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.

6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

<i>Team Member</i>	<i>Role(s)</i>	<i>Responsibilities</i>
Aubrey Carey	Project Manager	• Develop overall schedule for project • Runs meetings and makes overview slides • Reviews and monitors individual action items
Virginia Anderson	Lead Experimenter (ChemE); Sponsor Contact	• Lead person for design of experiments • Manages lab notebook and data collection • Establish contact with ChemTreat • Document meeting minutes
Rohan Sharma	Lead Developer (CS); Faculty Advisor Contact	• Manages the GitHub • Schedule meetings, send updates, and ensure deliverables are met for faculty advisors • Mentor team members by providing guidance and support to help team members improve their skills
Ronald Menendez	Financial Manager	• Research technical purchases and acquisitions • Conduct pricing analysis and budget justifications on proposed purchases • Carries out team purchase requests and monitors the budget

Step 5: Agree to the above team contract

Team Member: Aubrey Carey
Team Member: Virginia Anderson
Team Member: Rohan Sharma
Team Member: Ronald Menendez

Signature: Aubrey Carey
Signature: Virginia Anderson
Signature: Rohans
Signature: Ronald Menendez

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